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Université: Institut Universitaire des Sciences (IUS)

TD N° 9: Administration Systèmes Informatiques

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Niveau: 3^{ème} Année

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Mise en place d'un serveur mail sous Linux

Objectif :

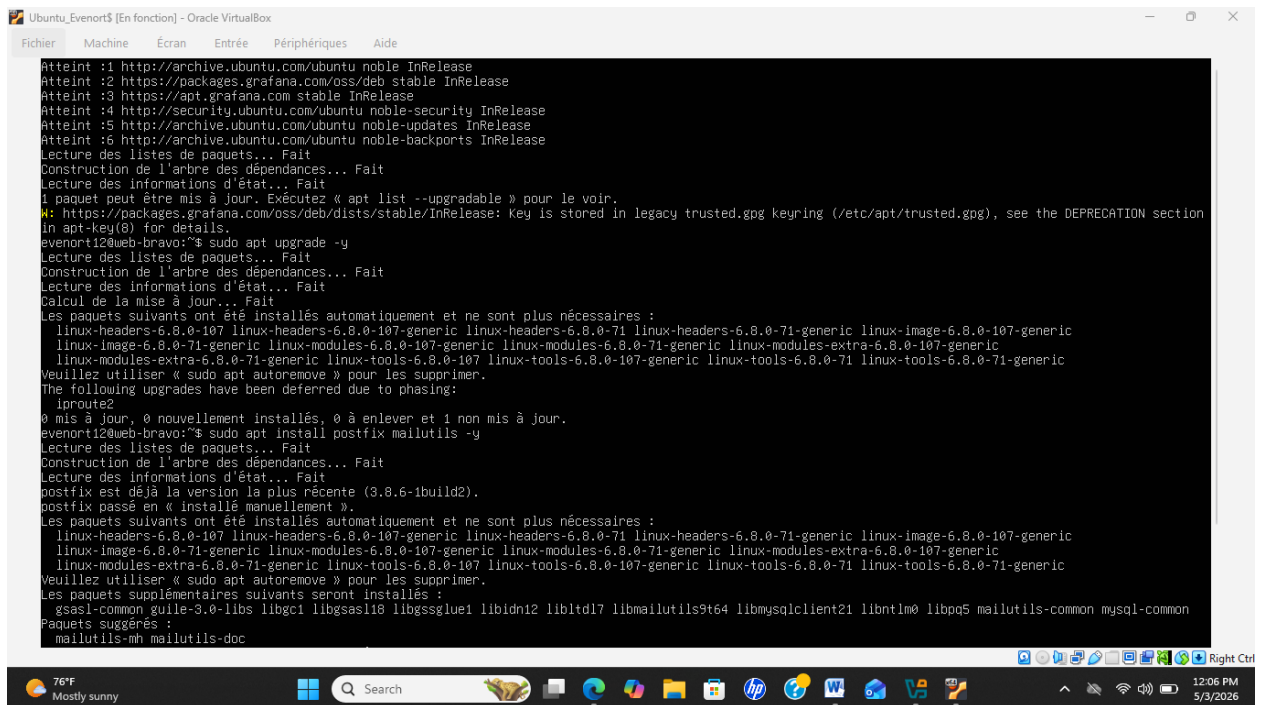
Installer et configurer un serveur mail fonctionnel avec :

- Postfix (SMTP)
- Dovecot (IMAP)
- Comptes utilisateurs
- Test en ligne de command

1. Préparation du système

`sudo apt update && sudo apt upgrade -y`

`sudo apt install postfix mailutils -y`



```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :2 https://packages.grafana.com/oss/deb stable InRelease
Atteint :3 https://apt.grafana.com stable InRelease
Atteint :4 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
1 paquet peut être mis à jour. Exécutez « apt list --upgradable » pour le voir.
W: https://packages.grafana.com/oss/deb/dists/stable/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section
in apt-key(8) for details.
evenort12@web-bravo:~$ sudo apt upgrade -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Calcul de la mise à jour... Fait
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
  linux-headers-6.8.0-107 linux-headers-6.8.0-107-generic linux-headers-6.8.0-71 linux-headers-6.8.0-71-generic linux-image-6.8.0-107-generic
  linux-image-6.8.0-71-generic linux-modules-6.8.0-107-generic linux-modules-6.8.0-71-generic linux-modules-extra-6.8.0-107-generic
  linux-modules-extra-6.8.0-71-generic linux-tools-6.8.0-107-generic linux-tools-6.8.0-71-generic linux-tools-6.8.0-71-generic
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
The following upgrades have been deferred due to phasing:
  iproute2
0 mis à jour, 0 nouvellement installés, 0 à enlever et 1 non mis à jour.
evenort12@web-bravo:~$ sudo apt install postfix mailutils -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
postfix est déjà la version la plus récente (3.8.6-1build2).
postfix passé en « installé manuellement ».
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
  linux-headers-6.8.0-107 linux-headers-6.8.0-107-generic linux-headers-6.8.0-71 linux-headers-6.8.0-71-generic linux-image-6.8.0-107-generic
  linux-image-6.8.0-71-generic linux-modules-6.8.0-107-generic linux-modules-6.8.0-71-generic linux-modules-extra-6.8.0-107-generic
  linux-modules-extra-6.8.0-71-generic linux-tools-6.8.0-107-generic linux-tools-6.8.0-71-generic linux-tools-6.8.0-71-generic
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
Les paquets supplémentaires suivants seront installés :
  gpg gpg-agent gpg-wks-client gpg-wks-server libgpg-error libgpgme libgpgme-bin libgpgme-dev libgpgme-doc libgpgme-glib
Paquets suggérés :
  mailutils-mh mailutils-doc
```

```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide

Paramétrage de libntlm0:amd64 (1.7-1build1) ...
Paramétrage de libidn12:amd64 (1.42-1build1) ...
Paramétrage de mailutils-common (1:3.17-1.1build3) ...
Paramétrage de libgssglue1:amd64 (0.9-1build1) ...
Paramétrage de libgc1:amd64 (1:8.2.6-1build1) ...
Paramétrage de libltdl7:amd64 (2.4.7-7build1) ...
Paramétrage de libgssasl18:amd64 (2.2.1-1willisync1build2) ...
Paramétrage de gsasl-common (2.2.1-1willisync1build2) ...
Paramétrage de guile-3.0-libs:amd64 (3.0.9-1build2) ...
Paramétrage de libmailutils9t64:amd64 (1:3.17-1.1build3) ...
Paramétrage de mailutils (1:3.17-1.1build3) ...
update-alternatives: utilisation de « /usr/bin/frm.mailutils » pour fournir « /usr/bin/frm » (frm) en mode automatique
update-alternatives: utilisation de « /usr/bin/from.mailutils » pour fournir « /usr/bin/from » (from) en mode automatique
update-alternatives: utilisation de « /usr/bin/messages.mailutils » pour fournir « /usr/bin/messages » (messages) en mode automatique
update-alternatives: utilisation de « /usr/bin/movemail.mailutils » pour fournir « /usr/bin/movemail » (movemail) en mode automatique
update-alternatives: utilisation de « /usr/bin/readmsg.mailutils » pour fournir « /usr/bin/readmsg » (readmsg) en mode automatique
update-alternatives: utilisation de « /usr/bin/dotlock.mailutils » pour fournir « /usr/bin/dotlock » (dotlock) en mode automatique
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
 6.8.0-118-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-111-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@web-bravo:~$ postconf mail_version
mail_version = 3.8.6
evenort12@web-bravo:~$
```

2. Configuration du fichier hosts

sudo nano /etc/hosts

Ajoute :

127.0.0.1 localhost

192.168.56.103 mail.ius.edu.ht mail

```
evenort12@web-bravo: ~
GNU nano 7.2 /etc/hosts *
127.0.0.1 localhost
127.0.1.1 web-bravo

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe80::0 ip6-localhost
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
127.0.0.1 localhost
192.168.56.103 mail.ius.edu.ht mail

Help Write Out Where Is Cut Paste Execute Location Undo Redo Set Mark To Bracket Previous
Exit Read File Replace Reuse Justify Go To Line Copy Copy Where Was Next
```

Test: ping mail.ius.edu.ht

3. Configuration de Postfix

sudo nano /etc/postfix/main.cf

Modifie/ajoute :

myhostname = mail.ius.edu.ht

mydomain = ius.edu.ht

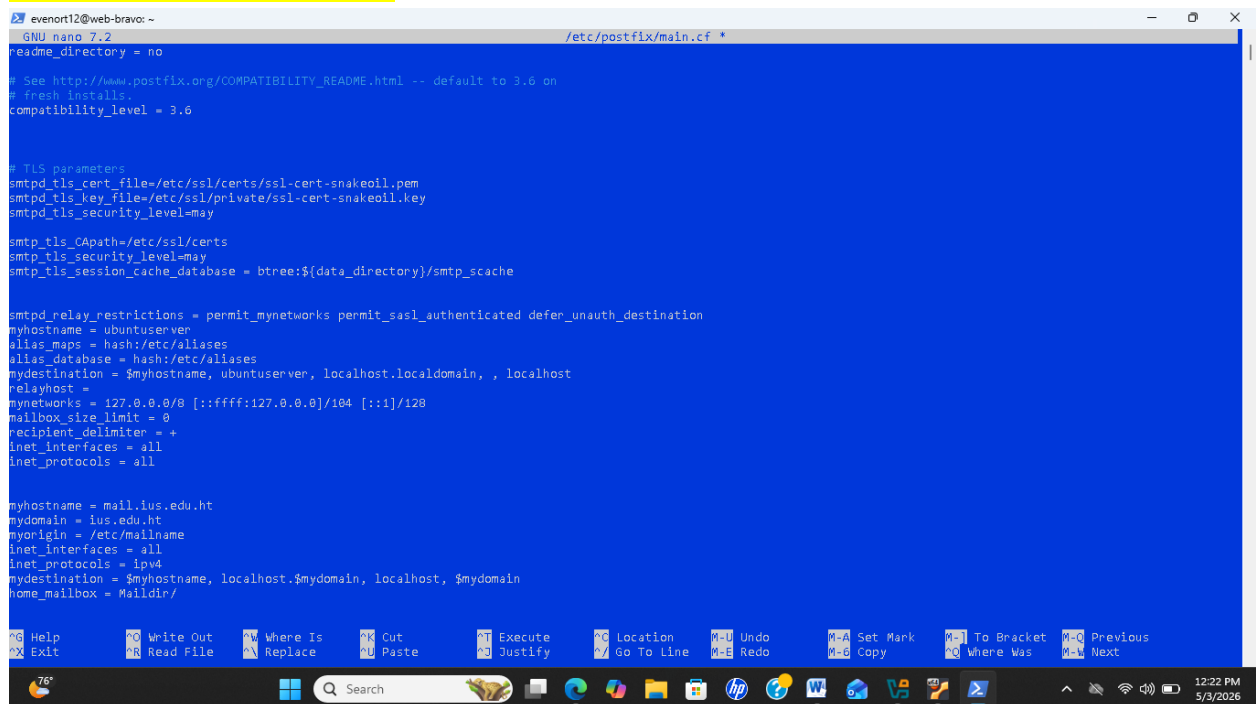
myorigin = /etc/mailname

inet_interfaces = all

inet_protocols = ipv4

mydestination = \$myhostname, localhost.\$mydomain, localhost, \$mydomain

home_mailbox = Maildir/



The screenshot shows a terminal window with the nano text editor open at the file /etc/postfix/main.cf. The editor's status bar at the top indicates 'GNU nano 7.2' and the file path. The configuration file content is as follows:

```
# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 3.6 on
# fresh installs.
compatibility_level = 3.6

# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtpd_tls_CApath=/etc/ssl/certs
smtpd_tls_security_level=may
smtpd_tls_session_cache_database = btree:${data_directory}/smtp_scache

smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = ubuntuserver
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = $myhostname, ubuntuserver, localhost.localdomain, , localhost
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all

myhostname = mail.ius.edu.ht
mydomain = ius.edu.ht
myorigin = /etc/mailname
inet_interfaces = all
inet_protocols = ipv4
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
home_mailbox = Maildir/
```

At the bottom of the terminal, there is a menu bar with various keyboard shortcuts for nano (e.g., ^G Help, ^X Exit, ^O Write Out, etc.). The system's taskbar is visible at the very bottom, showing the date and time as 12:22 PM on 5/3/2026.

4. J'ai redémarré : sudo systemctl restart postfix sudo systemctl status postfix

```
evenort12@web-bravo:~$ sudo nano /etc/hosts
evenort12@web-bravo:~$ sudo nano /etc/hosts
evenort12@web-bravo:~$ ping mail.ius.edu.ht
PING mail.ius.edu.ht (192.168.56.103) 56(84) bytes of data:
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=1 ttl=64 time=3.82 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=2 ttl=64 time=0.062 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=3 ttl=64 time=0.749 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=4 ttl=64 time=0.068 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=5 ttl=64 time=0.115 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=6 ttl=64 time=0.076 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=7 ttl=64 time=0.485 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=8 ttl=64 time=0.064 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=9 ttl=64 time=0.000 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=10 ttl=64 time=0.101 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=11 ttl=64 time=0.075 ms
64 bytes from mail.ius.edu.ht (192.168.56.103): icmp_seq=12 ttl=64 time=0.201 ms

--- mail.ius.edu.ht ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11450ms
rtt min/avg/max/mdev = 0.058/0.483/3.819/1.024 ms
evenort12@web-bravo:~$ sudo nano /etc/postfix/main.cf
evenort12@web-bravo:~$ sudo systemctl restart postfix
evenort12@web-bravo:~$ sudo systemctl status dovecot
Unit dovecot.service could not be found.
evenort12@web-bravo:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-03 16:23:29 UTC; 48s ago
     Docs: man:postfix(1)
   Process: 7932 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 7932 (code=exited, status=0/SUCCESS)
    CPU: 45ms

mai 03 16:23:29 web-bravo systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
mai 03 16:23:29 web-bravo systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
evenort12@web-bravo:~$
```

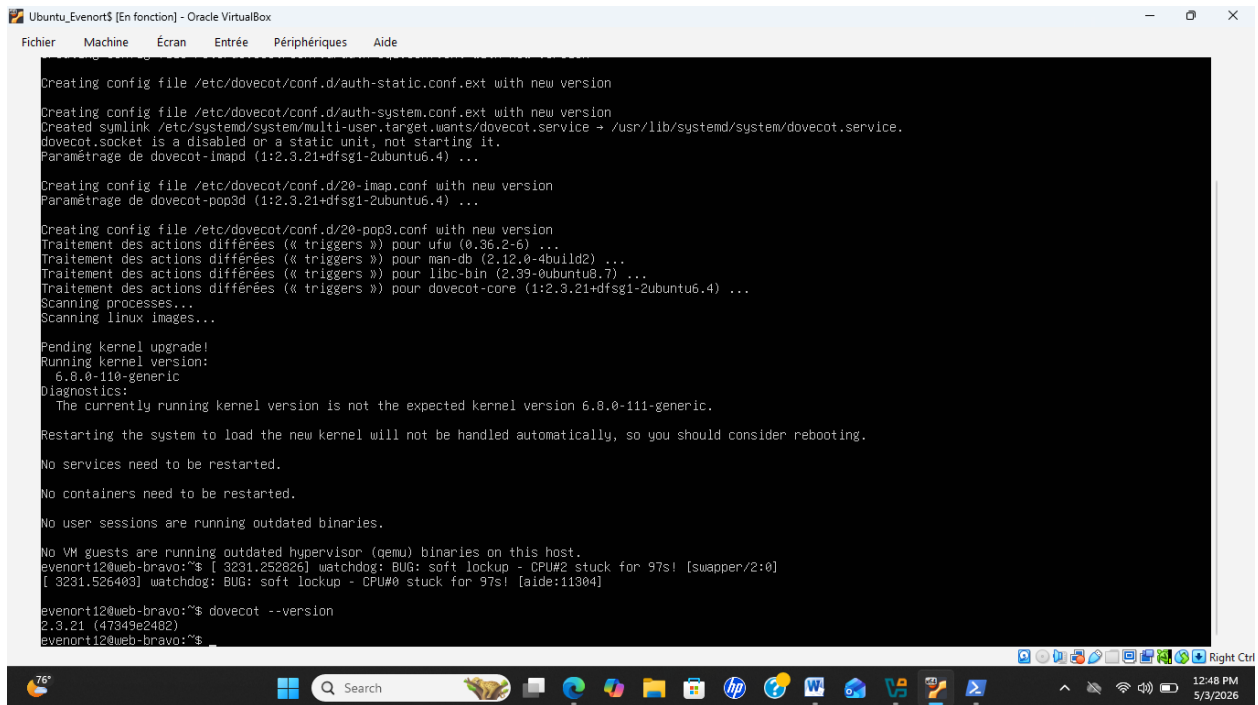
4.1 Installation des paquets

sudo apt install imapd dovecot-pop3d -y

Pendant l'installation :

Type : **Internet Site**

Domain : ius.edu.ht



```
Creating config file /etc/dovecot/conf.d/auth-static.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-system.conf.ext with new version
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.
dovecot.socket is a disabled or a static unit, not starting it.
Paramétrage de dovecot-imapd (1:2.3.21+dfsg1-2ubuntu6.4) ...
Creating config file /etc/dovecot/conf.d/20-imap.conf with new version
Paramétrage de dovecot-pop3d (1:2.3.21+dfsg1-2ubuntu6.4) ...
Creating config file /etc/dovecot/conf.d/20-pop3.conf with new version
Traitement des actions différées (« triggers ») pour ufw (0.36.2-6) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.39-0ubuntu8.7) ...
Traitement des actions différées (« triggers ») pour dovecot-core (1:2.3.21+dfsg1-2ubuntu6.4) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-110-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-111-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@web-bravo:~$ [ 3231.252826] watchdog: BUG: soft lockup - CPU#2 stuck for 97s! [swapper/2:0]
[ 3231.526403] watchdog: BUG: soft lockup - CPU#0 stuck for 97s! [aide:11304]

evenort12@web-bravo:~$ dovecot --version
2.3.21 (47349e2482)
evenort12@web-bravo:~$
```

5. Configuration de Dovecot

`sudo nano /etc/dovecot/dovecot.conf`

protocols : imap pop3

Maildir : `sudo nano /etc/dovecot/conf.d/10-mail.conf`

Ajout : `mail_location = maildir:~/Maildir`

5.1 J'ai redémarré: `sudo systemctl restart dovecot`

`sudo systemctl status dovecot`

```
evenort12@web-bravo: ~
GNU nano 7.2 /etc/dovecot/conf.d/10-mail.conf *
mail_location = maildir:~/Maildir

# Mail attachments
#
# sdbx and mbox support saving mail attachments to external files, which
# also allows single instance storage for them. Other backends don't support
# this for now.
# Directory root where to store mail attachments. Disabled, if empty.
#mail_attachment_dir =

# Attachments smaller than this aren't saved externally. It's also possible to
# write a plugin to disable saving specific attachments externally.
#mail_attachment_min_size = 128k

# Filesystem backend to use for saving attachments:
# posix : No SIS done by Dovecot (but this might help FS's own deduplication)
# sis posix : SIS with immediate byte-by-byte comparison during saving
# sis-queue posix : SIS with delayed comparison and deduplication
#mail_attachment_fs = sis posix

# Hash format to use in attachment filenames. You can add any text and
# variables: %(md4), %(md5), %(sha1), %(sha256), %(sha512), %(size).
# Variables can be truncated, e.g. %(sha256:80) returns only first 80 bits
#mail_attachment_hash = %(sha1)

# Settings to control adding $HasAttachment or $HasNoAttachment keywords.
# By default, all MIME parts with Content-Disposition=attachment, or inlines
# with filename parameter are considered attachments.
# add-flags - Add the keywords when saving new mails or when fetching can
# do it efficiently.
# content-type-type or ltype - Include/exclude content type. Excluding will
# never consider the matched MIME part as attachment. Including will only
# negate an exclusion (e.g. content-type=foo/* content-type=foo/bar).
# exclude-inlined - Exclude any Content-Disposition=inline MIME part.
#mail_attachment_detection_options =
mail_driver = maildir
mail_path = ~/Maildir

# Help
# Exit
# Write Out
# Read File
# Where Is
# Replace
# Cut
# Paste
# Execute
# Justify
# Location
# Go To Line
# Undo
# Redo
# Set Mark
# Copy
# To Bracket
# Where Was
# Previous
# Next
```

```
evenort12@web-bravo: ~$ sudo nano /etc/dovecot/conf.d/10-mail.conf
evenort12@web-bravo: ~$ sudo systemctl restart dovecot
evenort12@web-bravo: ~$ sudo systemctl status dovecot
* dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-03 16:58:29 UTC; 1min 18s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
   Main PID: 12935 (dovecot)
   Status: "v2.3.21 (47349e2482) running"
     Tasks: 4 (limit: 4455)
    Memory: 3.3M (peak: 3.5M)
       CPU: 297ms
   CGroup: /system.slice/dovecot.service
           └─12935 /usr/sbin/dovecot -F
             └─12936 dovecot/anvil
               └─12937 dovecot/log
                 └─12938 dovecot/config

mai 03 16:58:29 web-bravo systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
mai 03 16:58:29 web-bravo dovecot[12935]: master: Dovecot v2.3.21 (47349e2482) starting up for imap, pop3 (core dumps disabled)
mai 03 16:58:29 web-bravo systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.
evenort12@web-bravo: ~$
```

6. Création des utilisateurs

sudo adduser (jean, marie, luckson, kesnel, emma,

pierre, wilvick, luc, carl, woolems,

faudia, jude, gelin, cardo, fanie,

jacques, benson, edson, nelson, livens,

wideline, calixte, claude, wolly, jhon)


```
evenort12@web-bravo: ~
Info: Adding new user 'jean' to supplemental / extra groups 'users' ...
Info: Adding user 'jean' to group 'users' ...
evenort12@web-bravo: ~$ sudo adduser marie
Info: Adding user 'marie' ...
Info: Selecting UID/GID from range 1000 to 59999 ...
Info: Adding new group 'marie' (1005) ...
Info: Adding new user 'marie' (1005) with group 'marie (1005)' ...
Info: Creating home directory '/home/marie' ...
Info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for marie
Enter the new value, or press ENTER for the default
  Full Name []: marie Jeanne
    Room Number []: 78765656
    Work Phone []: 90988887
    Home Phone []: 45667667
    Other []:
Is the information correct? [Y/n] y
Info: Adding new user 'marie' to supplemental / extra groups 'users' ...
Info: Adding user 'marie' to group 'users' ...
evenort12@web-bravo: ~$ sudo adduser luckson
Info: Adding user 'luckson' ...
Info: Selecting UID/GID from range 1000 to 59999 ...
Info: Adding new group 'luckson' (1006) ...
Info: Adding new user 'luckson' (1006) with group 'luckson (1006)' ...
Info: Creating home directory '/home/luckson' ...
Info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for luckson
Enter the new value, or press ENTER for the default
  Full Name []: luckson louis
    Room Number []: 3
    Work Phone []: 45323455
    Home Phone []: 42354656
    Other []:
Is the information correct? [Y/n] y
Info: Adding new user 'luckson' to supplemental / extra groups 'users' ...
Info: Adding user 'luckson' to group 'users' ...
evenort12@web-bravo: ~$

evenort12@web-bravo: ~$
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109:/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:909:909:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
evenort12:x:1000:1000:Marc-Evenort:/home/evenort12:/bin/bash
sshd:x:109:65534::/run/ssh:/usr/sbin/nologin
postfix:x:110:111:/var/spool/postfix:/usr/sbin/nologin
aide:x:988:988:Advanced Intrusion Detection Environment:/var/lib/aide:/usr/sbin/nologin
prometheus:x:1001:1001:/home/prometheus:/bin/false
node_exporter:x:1002:1002:/home/node_exporter:/bin/false
node_exporter:x:1003:1003:/home/node_exporter:/bin/false
grafana:x:111:113:/usr/share/grafana:/bin/false
dovecot:x:112:114:Dovecot mail server,,,:/usr/lib/dovecot:/usr/sbin/nologin
dovnull:x:113:115:Dovecot login user,,,:/nonexistent:/usr/sbin/nologin
jean:x:1004:1004:jean lorens,1,4545665577,67678899,:/home/jean:/bin/bash
marie:x:1005:1005:marie Jeanne,78765656,90988887,45667667:/home/marie:/bin/bash
luckson:x:1006:1006:luckson louis,3,45323455,42354656:/home/luckson:/bin/bash
kesnel:x:1007:1007:kenel louis,4,43435522,464646487:/home/kesnel:/bin/bash
emma:x:1008:1008:emma pierre,5,34342534,23232456:/home/emma:/bin/bash
pierre:x:1009:1009:pierre marvens,6,32432546,32435343:/home/pierre:/bin/bash
wilwick:x:1010:1010:wilwick saint louis,32435676,94678965,544468893:/home/wilwick:/bin/bash
luc:x:1011:1011:luc thelusma,7,23234453,2223344:/home/luc:/bin/bash
carl:x:1012:1012:carl stanley,8,323445577,454566557:/home/carl:/bin/bash
woolems:x:1013:1013:woolems jean,9,123456754,213455678:/home/woolems:/bin/bash
faudia:x:1014:1014:faudia,10,124325566,234676655:/home/faudia:/bin/bash
jude:x:1015:1015:jude celestin,11,23244353,34253465:/home/jude:/bin/bash
gelin:x:1016:1016:gellin lesly,12,444534343,654765432:/home/gelin:/bin/bash
cardo:x:1017:1017:cardo dareus,13,1232345654,3245657686:/home/cardo:/bin/bash
fanie:x:1018:1018:fanie Jean louis,14,546576655,453456456:/home/fanie:/bin/bash
jacques:x:1019:1019:jacques,15,32434544,34256478:/home/jacques:/bin/bash
beneson:x:1020:1020:beneson honora,16,4323234645,3425346457:/home/benson:/bin/bash
edson:x:1021:1021:edson thelusma,17,223344353,231445356:/home/edson:/bin/bash
nelson:x:1022:1022:nelson louis,18,43434355,33445667:/home/nelson:/bin/bash
livens:x:1023:1023:livens duberiste,19,43544423,45365756:/home/livens:/bin/bash
wideline:x:1024:1024:wideline premier,20,33445423,33442234:/home/wideline:/bin/bash
calixte:x:1025:1025:calixte carolina,21,32324454,32435543:/home/calixte:/bin/bash
claudex:x:1026:1026:claudex jeune,22,434322332,345565443:/home/claudex:/bin/bash
wolly:x:1027:1027:wolly burfar,23,43546544,34234546:/home/wolly:/bin/bash
jhon:x:1028:1028:jhon nadir,24,432233554,445665343:/home/jhon:/bin/bash
evenort12@web-bravo: ~$
```

Professeurs (10)

sudo adduser (prof1, prof2, prof3, prof4, prof5,

prof6, prof7, prof8, prof9, prof10)

```
evenort12@web-bravo: ~  
$ sudo ./check_prof.sh  
uid=1029(prof1) gid=1029(prof1) groups=1029(prof1),100(users)  
uid=1030(prof2) gid=1030(prof2) groups=1030(prof2),100(users)  
uid=1031(prof3) gid=1031(prof3) groups=1031(prof3),100(users)  
uid=1032(prof4) gid=1032(prof4) groups=1032(prof4),100(users)  
uid=1033(prof5) gid=1033(prof5) groups=1033(prof5),100(users)  
uid=1034(prof6) gid=1034(prof6) groups=1034(prof6),100(users)  
uid=1035(prof7) gid=1035(prof7) groups=1035(prof7),100(users)  
uid=1036(prof8) gid=1036(prof8) groups=1036(prof8),100(users)  
uid=1037(prof9) gid=1037(prof9) groups=1037(prof9),100(users)  
uid=1038(prof10) gid=1038(prof10) groups=1038(prof10),100(users)  
evenort12@web-bravo: ~$
```

```
evenort12@web-bravo: ~  
GNU nano 7.2 check_prof.sh *  
#!/bin/bash  
  
domain="lus.edu.ht"  
  
echo " ^^^^ Verification des professeurs + emails..."  
  
for user in prof1 prof2 prof3 prof4 prof5 prof6 prof7 prof8 prof9 prof10  
do  
    echo "-----"  
    echo " ^^^ User: $user"  
  
    # Verify if user exists  
    if id "$user" >/dev/null 2>/dev/null  
    then  
        echo " ^^^t ^o Existe"  
  
        # ID Info  
        id $user  
  
        # Email  
        email="$user@$domain"  
        echo " ^^^s Email: $email"  
    else  
        echo " ^^^l N'existe pas"  
    fi  
done  
  
echo " ^^^^i V rification termin e"
```

```
evenort12@web-bravo: ~  
evenort12@web-bravo:~$ nano check_profs.sh  
evenort12@web-bravo:~$ chmod +x check_profs.sh  
sudo ./check_profs.sh  
# Verification des professeurs + emails...  
-----  
# User: prof1  
✓ Existe  
uid=1029(prof1) gid=1029(prof1) groups=1029(prof1),100(users)  
📧 Email: prof1@ius.edu.ht  
-----  
# User: prof2  
✓ Existe  
uid=1030(prof2) gid=1030(prof2) groups=1030(prof2),100(users)  
📧 Email: prof2@ius.edu.ht  
-----  
# User: prof3  
✓ Existe  
uid=1031(prof3) gid=1031(prof3) groups=1031(prof3),100(users)  
📧 Email: prof3@ius.edu.ht  
-----  
# User: prof4  
✓ Existe  
uid=1032(prof4) gid=1032(prof4) groups=1032(prof4),100(users)  
📧 Email: prof4@ius.edu.ht  
-----  
# User: prof5  
✓ Existe  
uid=1033(prof5) gid=1033(prof5) groups=1033(prof5),100(users)  
📧 Email: prof5@ius.edu.ht  
-----  
# User: prof6  
✓ Existe  
uid=1034(prof6) gid=1034(prof6) groups=1034(prof6),100(users)  
📧 Email: prof6@ius.edu.ht  
-----  
# User: prof7  
✓ Existe  
uid=1035(prof7) gid=1035(prof7) groups=1035(prof7),100(users)  
📧 Email: prof7@ius.edu.ht  
-----  
# User: prof8  
✓ Existe  
uid=1036(prof8) gid=1036(prof8) groups=1036(prof8),100(users)
```

Administratifs (5)

`sudo adduser (admin1, admin2, admin3, admin4, admin5)`

```
evenort12@web-bravo: ~  
evenort12@web-bravo:~$ for user in admin1 admin2 admin3 admin4 admin5  
do  
    echo "-----"  
    echo "# $user"  
    id $user  
done  
-----  
# admin1  
uid=1039(admin1) gid=1039(admin1) groups=1039(admin1),100(users)  
-----  
# admin2  
uid=1040(admin2) gid=1040(admin2) groups=1040(admin2),100(users)  
-----  
# admin3  
uid=1041(admin3) gid=1041(admin3) groups=1041(admin3),100(users)  
-----  
# admin4  
uid=1042(admin4) gid=1042(admin4) groups=1042(admin4),100(users)  
-----  
# admin5  
uid=1043(admin5) gid=1043(admin5) groups=1043(admin5),100(users)  
evenort12@web-bravo:~$
```

```
evenort12@web-bravo: ~
GNU nano 7.2          setup_admin.sh *
#!/bin/bash

domain "lus.edu.ht"

for user in admin1 admin2 admin3 admin4 admin5
do
    echo "-----"
    echo " ^q  Traitement de $user"

    # 1. Kreye user si li pa egziste
    id "$user" && /dev/null
    if [ $? -eq 0 ] ; then
        sudo adduser --disabled-password --gecos "" $user
        echo " ^|t  ^o Utilisateur cr "
    else
        echo " ^z  ^o D j  existant"
    fi

    # 2. Kreye Maildir
    mkdir -p /home/$user/Maildir/cur,new,tmp
    chown -R $user $user /home/$user/Maildir

    # 3. Email
    email "$user@$domain"
    echo " ^s  Email: $email"

    # 4. Affichage ID
    id $user
done

echo " ^n^1 Tous les administrateurs sont pr ts"
```

```
evenort12@web-bravo: ~
evenort12@web-bravo:~$ nano setup_admin.sh
evenort12@web-bravo:~$ chmod +x setup_admin.sh
evenort12@web-bravo:~$ sudo ./setup_admin.sh
[sudo] password for evenort12:
-----
! Traitement de admin1
^ Déjà existant
@ Email: admin1@lus.edu.ht
uid=1039(admin1) gid=1039(admin1) groups=1039(admin1),100(users)
-----
! Traitement de admin2
^ Déjà existant
@ Email: admin2@lus.edu.ht
uid=1040(admin2) gid=1040(admin2) groups=1040(admin2),100(users)
-----
! Traitement de admin3
^ Déjà existant
@ Email: admin3@lus.edu.ht
uid=1041(admin3) gid=1041(admin3) groups=1041(admin3),100(users)
-----
! Traitement de admin4
^ Déjà existant
@ Email: admin4@lus.edu.ht
uid=1042(admin4) gid=1042(admin4) groups=1042(admin4),100(users)
-----
! Traitement de admin5
^ Déjà existant
@ Email: admin5@lus.edu.ht
uid=1043(admin5) gid=1043(admin5) groups=1043(admin5),100(users)
# Tous les administrateurs sont prêts
evenort12@web-bravo:~$
```

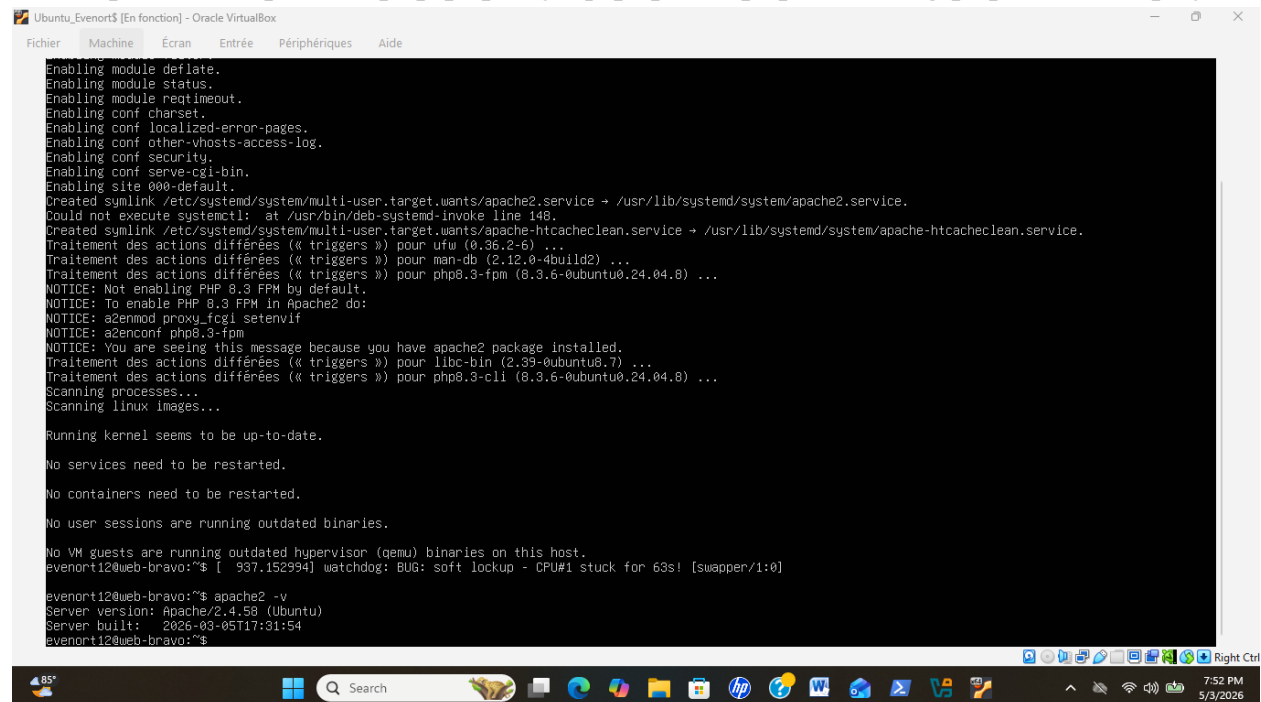
6.1 Maildir

```
evenort12@web-bravo: ~  
GNU nano 7.2 setup_mail.sh *  
#!/bin/bash  
  
domain "ius.edu.ht"  
  
users="jean marie luckson kesnel emma pierre wilvick luc carl woolems"  
faudia jude gelin cardo fanie jacques benson edson nelson livens  
wildeline calixte claudie wolly jhon"  
  
for user in $users  
do  
    echo " ^~ ^o Traitement de $user"  
  
    # 1. Kreye user si il pa egziste  
    id "$user" && /dev/null  
    if [ $? -eq 0 ]; then  
        sudo adduser --disabled-password --gecos "" $user  
    fi  
  
    # 2. Kreye email  
    email "$user@$domain"  
    echo " ^~^s Email: $email"  
  
    # 3. Kreye Maildir  
    sudo su $user mkdir -p /home/$user/Maildir/ cur,new,tmp  
  
    echo " ^|t ^o Termin pour $user"  
done  
  
echo " ^~^n^i Tous les utilisateurs sont pr ts"  
  
evenort12@web-bravo: ~  
evenort12@web-bravo:~$ nano setup_mail.sh  
evenort12@web-bravo:~$ chmod +x setup_mail.sh  
sudo ./setup_mail.sh  
[sudo] password for evenort12:  
Traitement de jean  
Email: jean@ius.edu.ht  
Terminé pour jean  
Traitement de marie  
Email: marie@ius.edu.ht  
Terminé pour marie  
Traitement de luckson  
Email: luckson@ius.edu.ht  
Terminé pour luckson  
Traitement de kesnel  
Email: kesnel@ius.edu.ht  
Terminé pour kesnel  
Traitement de emma  
Email: emma@ius.edu.ht  
Terminé pour emma  
Traitement de pierre  
Email: pierre@ius.edu.ht  
Terminé pour pierre  
Traitement de wilvick  
Email: wilvick@ius.edu.ht  
Terminé pour wilvick  
Traitement de luc  
Email: luc@ius.edu.ht  
Terminé pour luc  
Traitement de carl  
Email: carl@ius.edu.ht  
Terminé pour carl  
Traitement de woolems  
Email: woolems@ius.edu.ht  
Terminé pour woolems  
Traitement de faudia  
Email: faudia@ius.edu.ht  
Terminé pour faudia  
Traitement de jude  
Email: jude@ius.edu.ht  
Terminé pour jude  
Traitement de gelin  
Email: gelin@ius.edu.ht  
Terminé pour gelin
```

7. Installation de Roundcube (Webmail)

7.1 Installer Apache + PHP :

`sudo apt install apache2 php php-mysql php-intl php-mbstring php-xml unzip -y`



```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Could not execute systemctl: at /usr/bin/deb-systemd-invoke line 148.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Traitement des actions différées (« triggers ») pour ufw (0.36.2-6) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Traitement des actions différées (« triggers ») pour php8.3-fpm (8.3.6-0ubuntu0.24.04.8) ...
NOTICE: Not enabling PHP 8.3 FPM by default.
NOTICE: To enable PHP 8.3 FPM in Apache2 do:
NOTICE: a2enmod proxy_fcgi setenvif
NOTICE: a2enconf php8.3-fpm
NOTICE: You are seeing this message because you have apache2 package installed.
Traitement des actions différées (« triggers ») pour libc-bin (2.39-0ubuntu8.7) ...
Traitement des actions différées (« triggers ») pour php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

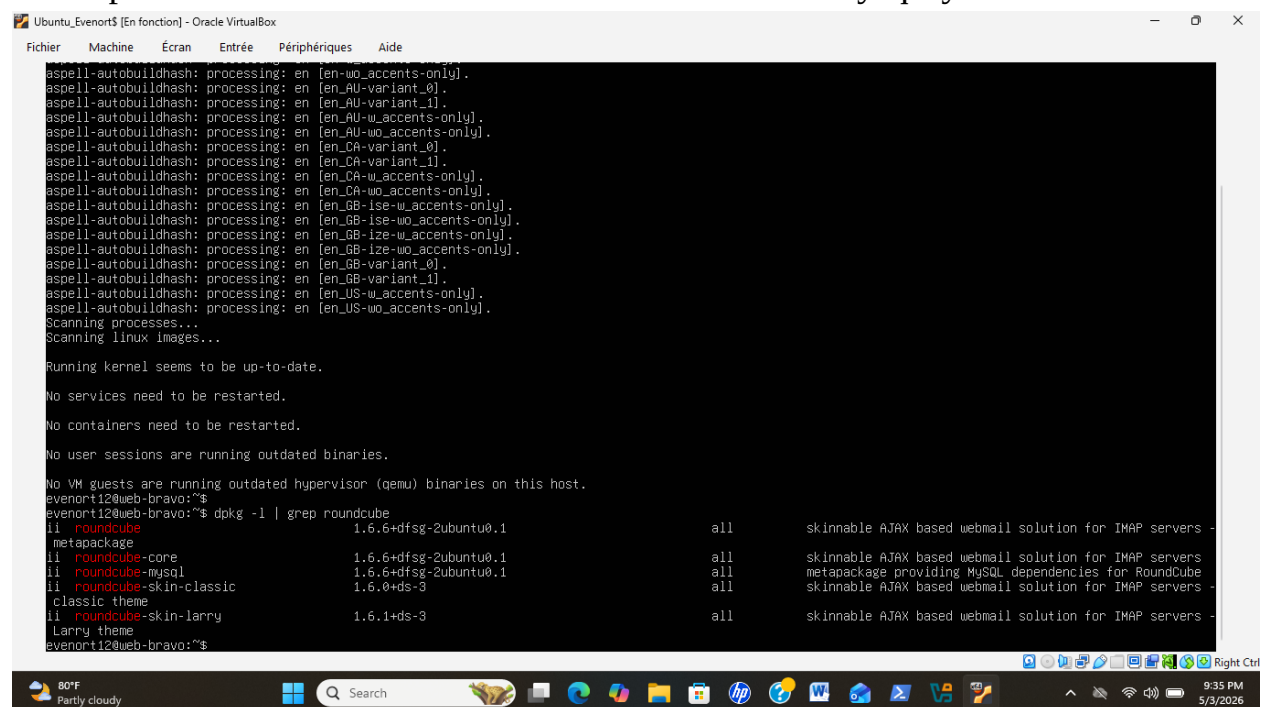
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@web-bravo:~$ [ 937.152994] watchdog: BUG: soft lockup - CPU#1 stuck for 63s! [swapper/1:0]

evenort12@web-bravo:~$ apache2 -v
Server version: Apache/2.4.58 (Ubuntu)
Server built: 2026-03-05T17:31:54
evenort12@web-bravo:~$
```

7.2 Installation Roundcube

`sudo apt install roundcube roundcube-core roundcube-mysql -y`



```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

aspell-autobuildhash: processing: en [en-wo_accents-only].
aspell-autobuildhash: processing: en [en_AU-variant_0].
aspell-autobuildhash: processing: en [en_AU-variant_1].
aspell-autobuildhash: processing: en [en_CA-variant_0].
aspell-autobuildhash: processing: en [en_CA-variant_1].
aspell-autobuildhash: processing: en [en_CA-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ise-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ise-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ize-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ize-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-variant_0].
aspell-autobuildhash: processing: en [en_GB-variant_1].
aspell-autobuildhash: processing: en [en_US-wo_accents-only].
aspell-autobuildhash: processing: en [en_US-wo_accents-only].
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

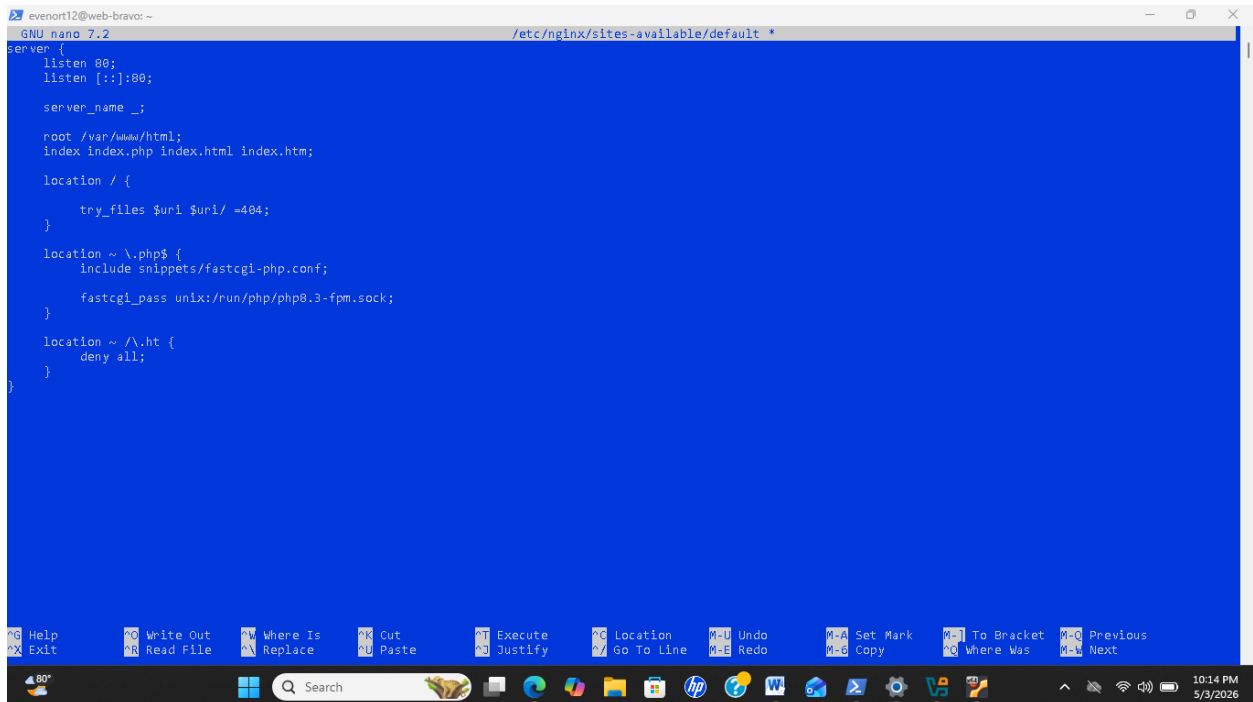
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@web-bravo:~$
evenort12@web-bravo:~$ dpkg -l | grep roundcube
ii roundcube               1.6.6+dfsg-2ubuntu0.1      all          skinnable AJAX based webmail solution for IMAP servers -
ii roundcube-core          1.6.6+dfsg-2ubuntu0.1      all          skinnable AJAX based webmail solution for IMAP servers
ii roundcube-mysql         1.6.6+dfsg-2ubuntu0.1      all          metapackage providing MySQL dependencies for RoundCube
ii roundcube-skin-classic  1.6.0+ds-3                 all          skinnable AJAX based webmail solution for IMAP servers -
ii roundcube-skin-larry    1.6.1+ds-3                 all          skinnable AJAX based webmail solution for IMAP servers -
evenort12@web-bravo:~$
```

7.3 Créer la configuration Nginx

`sudo nano /etc/nginx/sites-available/roundcube`



```
server {
    listen 80;
    listen [::]:80;

    server_name _;

    root /var/www/html;
    index index.php index.html index.htm;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
    }

    location ~ /\.ht {
        deny all;
    }
}
```

7.4 Activer le site

`sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/`

7.5 Supprime éventuellement le site par défaut :

`sudo rm /etc/nginx/sites-enabled/default`

7.6 Tester la config

`sudo nginx -t`

7.7 Redémarrer les services

`sudo systemctl restart php8.3-fpm`

`sudo systemctl restart nginx`

```
evenort12@web-bravo:~$ sudo nano /etc/nginx/sites-available/roundcube
[sudo] password for evenort12:
evenort12@web-bravo:~$ sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/
evenort12@web-bravo:~$ sudo rm /etc/nginx/sites-enabled/default
rm: cannot remove '/etc/nginx/sites-enabled/default': No such file or directory
evenort12@web-bravo:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
evenort12@web-bravo:~$ sudo systemctl restart php8.4-fpm
Failed to restart php8.4-fpm.service: Unit php8.4-fpm.service not found.
evenort12@web-bravo:~$ php -v
PHP 8.3.6 (cli) (built: Mar 20 2026 02:32:55) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
evenort12@web-bravo:~$ systemctl list-units --type=service | grep fpm
php8.3-fpm.service                                loaded active running The PHP 8.3 FastCGI Process Manager
evenort12@web-bravo:~$ ls /etc/php/
8.3
evenort12@web-bravo:~$ sudo systemctl restart php8.3-fpm
evenort12@web-bravo:~$ sudo systemctl restart nginx
evenort12@web-bravo:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-04 01:42:58 UTC; 55s ago
     Docs: man:nginx(8)
    Process: 4053 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=0)
    Process: 4056 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=0)
   Main PID: 4058 (nginx)
      Tasks: 2 (limit: 4455)
     Memory: 1.8M (peak: 2.0M)
        CPU: 85ms
    CGroup: /system.slice/nginx.service
            └─4058 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─4059 "nginx: worker process"

mai 04 01:42:58 web-bravo systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
mai 04 01:42:58 web-bravo systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
evenort12@web-bravo:~$

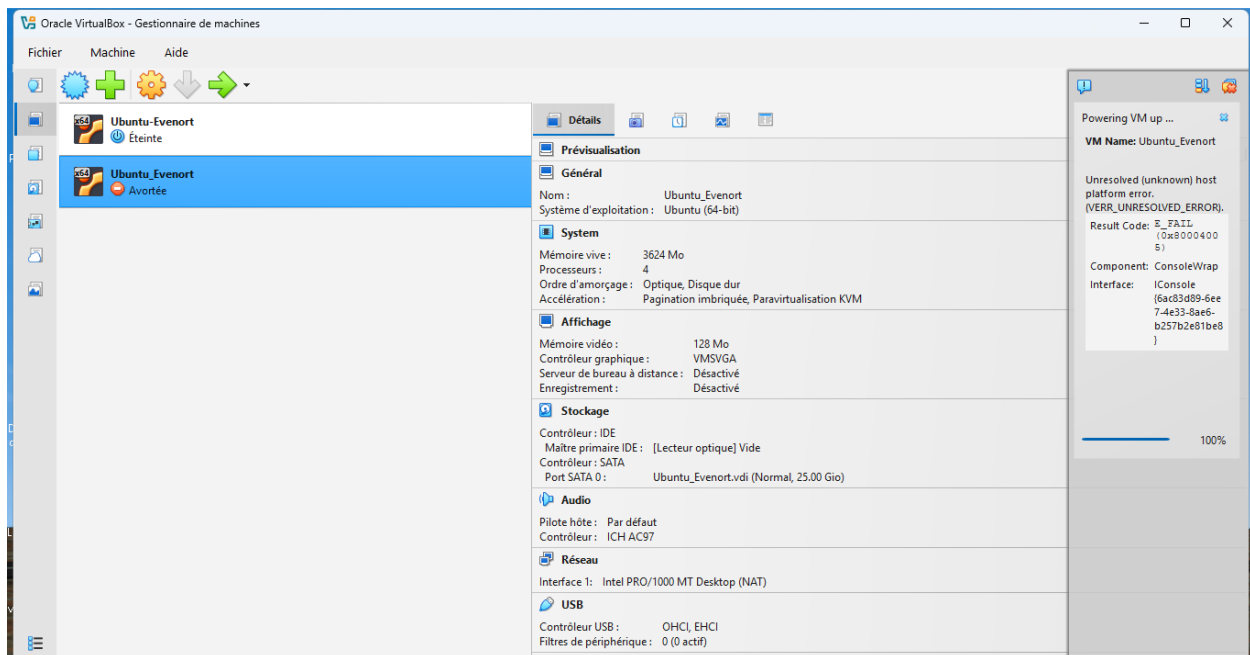
evenort12@web-bravo:~$ with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
evenort12@web-bravo:~$ systemctl list-units --type=service | grep fpm
php8.3-fpm.service                                loaded active running The PHP 8.3 FastCGI Process Manager
evenort12@web-bravo:~$ ls /etc/php/
8.3
evenort12@web-bravo:~$ sudo systemctl restart php8.3-fpm
evenort12@web-bravo:~$ sudo systemctl restart nginx
evenort12@web-bravo:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-04 01:42:58 UTC; 55s ago
     Docs: man:nginx(8)
    Process: 4053 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=0)
    Process: 4056 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=0)
   Main PID: 4058 (nginx)
      Tasks: 2 (limit: 4455)
     Memory: 1.8M (peak: 2.0M)
        CPU: 85ms
    CGroup: /system.slice/nginx.service
            └─4058 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─4059 "nginx: worker process"

mai 04 01:42:58 web-bravo systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
mai 04 01:42:58 web-bravo systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
evenort12@web-bravo:~$
evenort12@web-bravo:~$ php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-04 01:42:50 UTC; 2min 3s ago
     Docs: man:php-fpm8.3(8)
    Process: 4042 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install --run/php/php-fpm.sock --etc/php/8.3/fpm/pool.d/www.conf 03 (code=exited, status=0/SUCCESS)
   Main PID: 4039 (php-fpm8.3)
    Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"
      Tasks: 3 (limit: 4455)
     Memory: 0.7M (peak: 10.7M)
        CPU: 338ms
    CGroup: /system.slice/php8.3-fpm.service
            └─4039 "php-fpm: master process (/etc/php/8.3/fpm/php-fpm.conf)"
               └─4040 "php-fpm: pool www"
                  └─4041 "php-fpm: pool www"

mai 04 01:42:49 web-bravo systemd[1]: Starting php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager...
mai 04 01:42:50 web-bravo systemd[1]: Started php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager.
evenort12@web-bravo:~$
```

Accès

<http://192.168.56.103/roundcube>



Remarque : Au cours de la réalisation de ce projet, une panne technique a provoqué la perte du serveur Linux utilisé pour les tests. En raison du temps limité avant la date de remise, il n'a pas été possible de refaire l'ensemble des configurations et des validations. Le devoir est donc soumis dans son état actuel, avec les éléments qui ont pu être réalisés avant l'incident.

Connexion

Utilisateur : support

Mot de passe : *****

Modifier /etc/hosts

sudo nano /etc/hosts

192.168.56.103 mail.devcloud.local mail

sudo hostnamectl set-hostname mail.devcloud.local

hostname -f

Tests fonctionnels

Vérification depuis le terminal (mail)

su – admin

8. Vérifier si le dossier Maildir existe

Connecté en admin, tape : `ls -la ~`

Ex : etudiant1 → **etudiant1@ius.edu.ht**

prof1 → prof1@ius.edu.ht

8.1 Test en ligne de commande (local)

Envoyer un mail :

```
echo "Bonjour test mail" | mail -s "Test" etudiant1
```

8.2 Installer mail

```
sudo apt install mailutils -y
```

Lire les mails : `su - etudiant1`

Mail

8.3 Vérification des services

```
sudo systemctl status dovecot
```

9. Tester les ports

```
ss -tlnp
```

10. (Bonus) Activer firewall

```
sudo ufw allow 25
```

```
sudo ufw allow 143
```

```
sudo ufw allow 110
```

```
sudo ufw enable
```

Conclusion et Rapport d'Incident Technique

La mise en place du serveur de messagerie de l'IUS Haïti a permis de configurer avec succès les services **Postfix et Dovecot** pour le domaine ius haiti.edu.

Cependant, une panne système majeure est survenue durant les phases finales de test, empêchant l'achèvement complet du projet. Malgré cette difficulté, les principaux objectifs de configuration ont été atteints. Une réinstallation du système et une meilleure gestion de la base de données seront nécessaires afin de finaliser les tests restants et assurer le bon fonctionnement du serveur de messagerie.